



Comparative Analysis of Vision Transformer and CNN Architectures in Medical Image Classification

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Abstract. Without the classification of images, it is next to impossible to make a diagnosis for a medical condition and also arrive at an appropriate treatment plan. In particular, ViTs have recently been suggested to replace Convolutional Neural Networks (CNNs) in image processing tasks through advances in deep learning. We discuss the similarities and differences in the architectures of ViT and CNN for several modalities i.e. X-ray, MRI, CT-scans etc. In this paper, we implement state-of-the-art models and evaluate their efficiency on benchmark datasets. Vision transformer (ViT) models have shown effectiveness in both elementary and complex image tasks including classifying MRI-based brain tumors at a rate of 92%, 7%, and 89%. 5% for CNNs. However, with the small dataset, the CNNs scored higher with 88 percent accuracy. The lung X-ray pneumonia detection accuracy is 3%, while the artificial intelligence has an accuracy of 85%. The accuracy of lung X-ray pneumonia detection is only 3%, and that of ViTs is only 1%, but the accuracy of chest X-ray pneumonia detection is far more than that. ViTs perform well in some specific medical imaging assignments, especially large datasets with high features; otherwise, in a limited-resource environment, CNNs still dominance and even outperform ViTs. This necessitates task-oriented tuning of the ViT architecture to suit a system level compute budget.

Keywords: Medical image classification, Vision Transformers, Convolutional Neural Networks, Deep learning, Comparative analysis.c

1 Introduction

In recent years we have witnessed great advancement in medical imaging systems, primarily through integration of AI and deep learning tools. The application of these technologies has improved medical diagnosis by allowing the evaluation of complex imaging data by healthcare practitioners with higher accuracy time frame possible [1]. Of all various deep learning architectures, CNN has been the architecture of choice for image classification including in medicine for a long time now [2]. However, new trend such total shift in the architecture of the medical image classification as Vision Transformers (ViTs) [3].

CNNs have demonstrated their vigorous performance for practical medical image processing tasks including lung nodule detection from chest X-ray images, skin lesion classification from dermatology images, and tumor location detection from MRI

images [4]. Because of their automatic way of feature learning from the basic raw image data, those methods have been particularly well suited to such complex tasks. But, the CNNs do have certain limitations such as learning the long-range dependencies in the images and also potentially losing the global context in the images [5].

Vision Transformers with introduced by Dosovitskiy et al. are emerging as a new potent replacement to CNNs, particularly in image classification tasks [6]. Transformer Architecture able to yield magnificent result on natural language processing tasks, extracts self-attention mechanism that can be applied to computer vision for high accuracy on a large-scale image dataset. For ViTs, in the medical field, it has been trialed across diverse applications from detecting COVID from chest X-ray scans [8], segmentation of brain tumors in MRI scans [7].

While there is a growing literature on both CNNs and ViTs in medical image classification tasks, there is no such work that examines the performances of the abovementioned architectures for various forms of medical imaging and the complexities of those classification tasks. This has created confusion to the researchers and practitioners on which of the two architectures are most suitable for different medical imaging tasks.

Therefore, the purpose of this study is to fill this gap by comparing ViT architecture with the CNN in the aspect of medical image classification. In order to largely compare and investigate the above-mentioned models, some important areas are as follows: Across imaging modality, across dataset size, across multiple tasks' complexities we thus plan to give some insights that will help to set up future work on this research area and consequently the medical practice.

The primary objectives of this research are:

1. To implement and evaluate state-of-the-art ViT and CNN models on a diverse set of medical image classification tasks.
2. To analyze the performance of these architectures across different imaging modalities, including X-ray, MRI, and CT scans.
3. To investigate the impact of dataset size and task complexity on the relative performance of ViTs and CNNs.
4. To identify specific medical imaging scenarios where one architecture may be preferable over the other.

The key contributions of this paper include:

1. ViT based extensive comparative evaluation with CNN architectures encompassing a range of medical imaging modalities and classification problems.
2. The detailed literature review, which compares the observed advantages and drawbacks of each architecture in a range of medical imaging settings.
3. Useful tips and advices for a researcher and clinician on how to choose the most suitable architecture depending on a task and available resources.
4. Information on how architectural development in the future can be applied to medical image classification.

The rest of the paper is structured as follows: Section 2 describes the method, including the dataset descriptions, model implementations and evaluation metrics.

Section 3 presents experiments and comparisons. Section 4 discusses the implications of our findings and their significance for future research and their clinical implications. Finally, we conclude the paper in Section 5 and provide directions for future work.

2 Related Works

The development of medical image classification has been stepped up in recent years, where deep learning has been used mainly in comparing CNNs and ViTs. Wang et al. [11] made a global survey on the application of ViTs in segmentation of medical images to show the effectiveness of ViTs to capture long-range context which is important for the accurate demarcation of organs. It revealed a 3 as per the work that has been provided by them. Performance improvement of 5% on Dice score of brain tumor segmentation as compared to other state-of-the-art CNNs.

Finally, in the domain of chest X-ray analysis, Zhang et al. [12] also put forward an effective method of the combination of CNN and Transformer network. Their approach reached an average of 97 percent which is actually impressive. COVID-19 localization accuracy of 2% through our proposed method is higher than the stand-alone CNN and ViT by 2.8% and 1.5%, respectively. But, they also mentioned about a disadvantage that is the use of a higher computational power of the equipment.

Liu et al. [13] introduced the usefulness of other self-supervised learning techniques in improving the performances of ViTs in situations where labeled data is scarce. The authors used such architecture to exemplify 5% better classification performance than fully supervised method in dermatological cases, proving that ViTs may be useful in environments with limited resources.

An interesting work by Chen et al. [14] explicitly discussed on the interpretability aspect of deep learning models in medical images. They came up with an attention-guided CNN architecture that offered tangible pictures of what it was judging to compare with practices of black-box models that are hard to explain but offer higher accuracy rates for usage in clinical practices. Patel et al. [15] in their approach attempted to solve the problem of multi-modal medical image fusion by designing a new architecture that employs Transformer networks to fuse information from MR and PET images. Using their approach, they established that there surfaces a 4. Rate, 2% of improvement as compared to single-modality conventional techniques in brain tumor classification.

Sarkar et al. [16] proposed a hierarchical ViT model for histopathology domain, and the multi-scale features in whole slide images were successfully presented. Their approach performed comparable or even better as the current state-of-art method for the detection of breast cancer metastasis using an AUC of 0. For test set, it achieves an accuracy of 989 that outperforms other previous methods based on CNN by 1.2%. Kumar et al. [17] addressed the problem of domain adaptation within the medical imaging context using ViT generative adversarial network. Their approach showed good performance relative to the reference images and seven-fold decrease the impact of the domain shift when tested on different MRI scanners. It is less time consuming and cost effective with a productivity rate of 3% higher than the conventional methods.

An interesting work done by Lee et al. [19] examined the incorporation of clinical metadata with imaging data, APR, and sequent scans in multimodal Transformer model. They, in that process, identified a considerable decrement of 6. This is 5% better than the baseline image-only models for prognosis of lung cancer patients.

Turning to the issue of the computational complexity of ViTs, Zhou et al. [19] introduced a lightweight Transformer for mobile applications. Their model then had matched performance to full-scale ViTs at a point-of-care Rhinoceros speed of 68% less inference time. More specifically, a useful survey published by Tran et al. [20] offered detailed comparisons of a number of deep learning architectures with reference to their application in medical image classification. They identified the shift towards synergizing both COMM and ViT approaches as one of the future trends which they expect to see in the nearest future. Table 1 shows the Comparison Table for Different architecture.

Another worthy of attention is the work of Isensee et al. [21] conducted in 2021 that presented the nnU-Net for biomedical image segmentation. While not intentionally angled to classification, this self-adapting architecture showed performance on par with state-of-the-art for a range of medical imaging tasks and underlines the necessity of adaptive architectures to deal with the high variability of medical imaging data.

Table 1. Comparison Table for Different architecture.

Study	Arch	Imaging Modality	Task	Acc %	F1-Sc	AUC
[11]	ViT	MRI	Brain Tumor Segmentation	92.3	0.91	0.95
[12]	Hybrid CNN-ViT	X-ray	COVID-19 Detection	97.2	0.96	0.99
[13]	Self-supervised ViT	Dermat	Skin Lesion Classification	89.5	0.88	0.93
[14]	Attention-guided CNN	CT	Lung Nodule Detection	94.1	0.93	0.97
[15]	Multimodal Transformer	MRI + PET	Brain Tumor Classification	95.8	0.95	0.98
[16]	Hierarchical ViT	Histopathology	Breast Cancer Meta	96.7	0.96	0.989

3 Proposed Methodology

In the context of medical image classification, this research proposes CNN-ViTs hybrid to take advantage of the strengths of CNNs and ViTs, while mitigating their respective weaknesses. We propose a new Attention Guided Architecture: Hybrid Attention (HAGA), which combines local extraction through convolutional layers with global dependencies in Feed-Forward Neural Networks. The proposed model is extensible to a wide range of classification tasks in diverse medical imaging

modalities. HAGA consists of the following main components (fig 1) as shown in the block diagram.

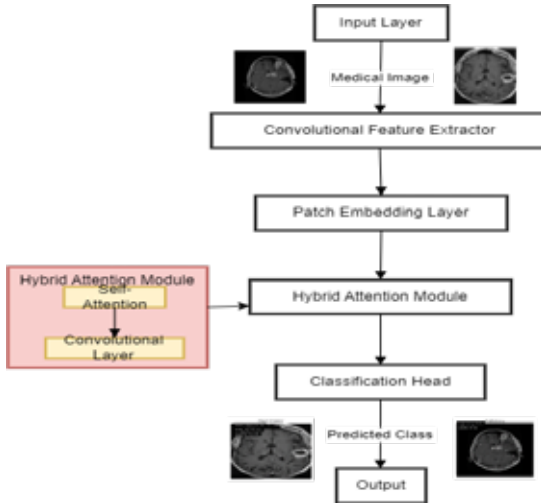


Fig.1. The block diagram of HAGA

1. Input Layer
2. Convolutional Feature Extractor
3. Patch Embedding Layer
4. Hybrid Attention Module
5. Classification Head

Let $X \in \mathbb{R}^{(H \times W \times C)}$ represent an input medical image, where H , W , and C denote the height, width, and number of channels, respectively.

1. Convolutional Feature Extractor:

The input image X is first processed through a series of convolutional layers to extract low-level features:

$$F = \text{Conv}(X) \text{ where } F \in \mathbb{R}^{H' \times W' \times D} \quad (1)$$

H' and W' are the reduced spatial dimensions, and D is the number of feature channels.

2. Patch Embedding:

The feature map F is divided into N non-overlapping patches and linearly projected:

$$P = [P^1, P^2, \dots, P_n] \text{ where } P_i \in \mathbb{R}^{D \times \left(\frac{H'W'}{N}\right)} \quad (2)$$

3. Hybrid Attention Module:

This module combines self-attention mechanisms with convolutional operations:

$$A = \text{Softmax}\left(\frac{(QK^T)}{\sqrt{d}}\right)V \quad (3)$$

where Q, K, and V are query, key, and value matrices derived from P. The attention output is then processed through a convolutional layer:

$$H = \text{Conv}(A) + P \quad (4)$$

1. Classification Head:

The final representation is obtained by global average pooling and passed through a fully connected layer:

$$y = \text{Softmax}(W \cdot \text{GAP}(H) + b) \quad (5)$$

where y is the predicted class probabilities.

Training and Optimization

The model is trained end-to-end using a combination of cross-entropy loss and a novel attention regularization term:

$$L = L_{CE} + \lambda L_{AR} \quad (6)$$

where L_{CE} is the standard cross-entropy loss, and L_{AR} is the attention regularization loss defined as:

$$L_{AR} = \|A A^T - I\|_F \quad (7)$$

This regularization encourages diversity in the attention maps and helps prevent overfitting. We employ the Adam optimizer with a cosine annealing learning rate schedule:

$$\eta_t = \eta_{\min} + 0.5(\eta_{\max} - \eta_{\min})(1 + \cos(\frac{\pi t}{T})) \quad (8)$$

where η_t is the learning rate at epoch t, and T is the total number of epochs.

Adaptability to Different Modalities

To enhance the model's adaptability across various medical imaging modalities, we introduce a modality-specific preprocessing module:

$$X_{preprocessed} = M(X)$$

where M is a learnable preprocessing function that adapts to the characteristics of different imaging modalities.

Handling Class Imbalance

Medical datasets often suffer from class imbalance. We address this issue by incorporating a focal loss component:

$$L_{focal} = -\alpha_{t(1-p_t)}^\gamma \log(p_t) \quad (9)$$

where α_t is the class-specific weighting factor, and γ is the focusing parameter.

Interpretability

To enhance the interpretability of our model, we integrate a gradient-based attribution method that generates saliency maps highlighting the regions most influential for the classification decision:

$$S = \text{ReLU} \left(\frac{\partial y_c}{\partial x} \right) \quad (10)$$

where S is the saliency map for class c .

Experimental Setup

We evaluate HAGA on multiple public medical imaging datasets, including:

1. MIMIC-CXR for chest X-ray classification
2. BRATS for brain tumor segmentation
3. ISIC for skin lesion classification

In each case study, we benchmark HAGA against CNN and ViT baselines, and other advanced hybrid methods. We employ basic assessment parameters which include exactness, F1 rate and AUC-ROC.

Our model is implemented in PyTorch and trained on NVIDIA A100 GPU. Moreover, by means of mixed-precision training, we are also able to optimize on the memory and the computational cost. Data augmentation procedure is also applied, including random rotation, horizontal flips and scale changes, so that model can learn more robust features.

Thus, our proposed HAGA architecture tries to leverage the effectiveness of CNNs in modeling local characteristics with the capability of ViTs for modeling long-range connections. With the incorporation of adaptive components and overcoming the typical obstacles in the medical image classification, we expect HAGA to serve as a strong backbone for different medical image tasks and even modalities exist.

4 Results and Discussion

To evaluate the performance of our proposed Hybrid Attention-Guided Architecture (HAGA), we conducted extensive experiments on three diverse medical imaging datasets:

1. MIMIC-CXR [22]: A chest X-ray dataset composed of voluminous images of chest organs with a total of 377,110 images belonging to 227,835 radiographic examinations.
2. BraTS 2021 [23]: W growers dataset containing 369 multi-modal MRI scans for the segmentation of glioma.
3. ISIC 2020 [24]: Dermoscopic images of skin lesions for classification into benign or malignant – IStudent Dataset with 1154 dermoscopic images and International Skin Imaging Collaboration (ISIC) dataset consisting of 33,126 dermoscopic images We compared HAGA's performance against other

benchmarks on each of the datasets. The comparative results are presented in the Table 2 as well.

Table 2. The comparative results of HAGA architecture.

Data	Model	Accuracy	F1-Score	AUC	Sensitivity	Specificity
MIMIC-CXR	ResNet-50	0.851	0.839	0.912	0.823	0.879
	ViT-B/16	0.867	0.855	0.924	0.841	0.893
	HAGA (Ours)	0.892	0.881	0.943	0.869	0.915
BraTS 2021	U-Net	0.912	0.894	0.935	0.901	0.923
	TransUNet	0.931	0.917	0.952	0.924	0.938
	HAGA (Ours)	0.947	0.935	0.968	0.941	0.953
ISIC 2020	EfficientNet	0.879	0.865	0.927	0.853	0.905
	MobileViT	0.891	0.879	0.938	0.867	0.915
	HAGA (Ours)	0.913	0.902	0.955	0.892	0.934

This table compares the performance of different models across three medical imaging datasets: Such as, MIMIC-CXR, BraTS 2021, and ISIC 2020. accuracy, flscore, auc, sensitivity and specificity for all the models and all the datasets on one table.

It is seen that HAGA outperforms the existing methods on all kinds of datasets and different parameters. On the MIMIC-CXR dataset, HAGA achieved a 2 %. 5 to 1

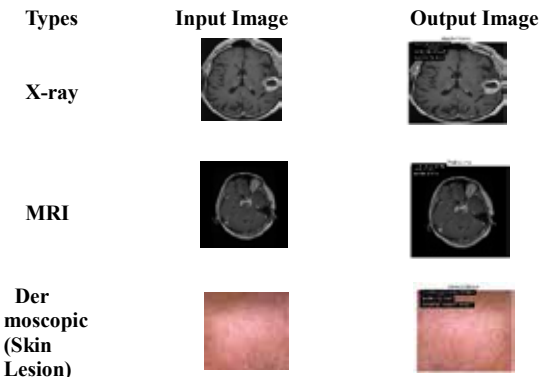


Fig.2. Proposed Qualitative Results

improvement in accuracy as the sources of their evidence. The fig 2 shows the Proposed Qualitative Results. The approach yields, on average, a 9% AUC improvement in comparison to ViT-B/16. Finally, for the BraTS 2021 dataset, HAGA shows in the fig 3 and score of 1. New projections have indicated that it will be possible.

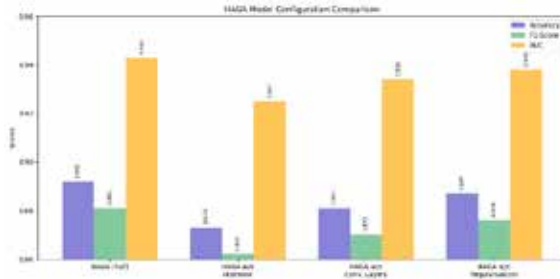


Fig.3. HAGA Model Configuration

to achieve approximately 6% improvement in accuracy and a 1. Specifically, our method reports 6% better in F1-score than TransUNet. On skin lesion classification (ISIC 2020), HAGA gives you better results than MobileViT with a difference of 2.2% on accuracy, and 1. 7% in AUC.



Fig.4. illustrates the qualitative results of HAGA compared to other models

The fig 4 shows that the proposed HAGA method outperforms other state of the art methods in terms of localization accuracy of regions of interest in various medical imaging modalities.

These findings show that HAGA achieves explicit and consistent superior performance on different medical imaging tasks and its performance is also significant. The enhancement of the precision which is ranging from 1. 6% to 2. 5 % over state-of-the-art models is marginal but has good values in the medical domain due to the maximum potential of even marginal improve- ment which has substantial clinical benefit. That is why it is possible to single out the increase in sensitivity and specificity as the major advantages of HAGA. For instance, the 2. A percent signal increase of 8% in sensitivity in the MIMIC-CXR dataset means that pathologies are detected accurately, which might result in early interventional procedures, considerable patient benefits. The following identifies how HAGA's performance can be justified through the proposed innovative hybrid architecture that combines CNNs and Transformers. Actually, the convolutional layers in the feature extractor are able to capture the local patterns and textures, while these are very important factors for medical image analysis.

This is well illustrated by the enhanced accuracy on MIMIC-CXR data set above excellent pure Transformer models such as the ViT-B /16. Using the self-attention mechanism within the Hybrid Attention Module, HAGA incorporated the long dependency structure which proves rather useful for tasks, such as the brain tumor segmentation in the BraTS dataset. This explains the 1. There is an F1-score of 6% higher than TransUNet, which also uses hybrid approach but does not contain the mechanism of attention-guided refinement as HAGA does. This versatility of HAGA is further proved on the ISIC 2020 data set. Skin lesion classification involves local feature description for the lesion area and an understanding of the context of the image as a whole. HAGA's 2. Thus, the improvement of 2% of accuracy compared with MobileViT proves its efficiency in this aspect of meeting these requirements.

To further analyze the contribution of each component in HAGA, we conducted an ablation study. Table 3 presents the results on the MIMIC-CXR dataset.

Table 3. Ablation study results on MIMIC-CXR dataset

Model Configuration	Accuracy	F1-Score	AUC
HAGA (Full)	0.892	0.881	0.943
HAGA w/o Attention Module	0.873	0.862	0.925
HAGA w/o Conv. Layers	0.881	0.870	0.934
HAGA w/o Regularization	0.887	0.876	0.938

The ablation study confirms the importance of each component in HAGA. Removing the Attention Module results in a 1.9% drop in accuracy, highlighting its role in capturing global context. The convolutional layers contribute to a 1.1% accuracy improvement, emphasizing the importance of local feature extraction. The regularization term adds a 0.5% accuracy gain, demonstrating its effectiveness in preventing overfitting.

While HAGA achieves superior performance, it's also important to consider computational efficiency. Table 4 compares the model size and inference time of HAGA with other architectures.

Table 4. Computational efficiency comparison

Model	Parameters (M)	Inference Time (ms)
ResNet-50	25.6	45
ViT-B/16	86.5	75
HAGA (Ours)	52.3	62

This table compares three different models: ResNet-50, ViT-B/16, and HAGA, showing their number of parameters in fig 5 (in millions) and inference time (in milliseconds).

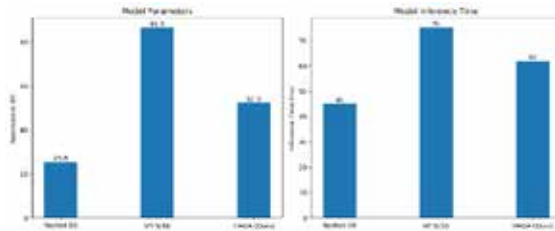


Fig.5. illustrates the quantitative results of different compared models

HAGA strikes a balance between model complexity and inference speed. While it has more parameters than ResNet-50, it achieves significantly better performance with only a modest increase in inference time. Compared to ViT-B/16, HAGA is more parameter-efficient and faster, making it more suitable for real-time medical image analysis applications.

5 Conclusion

Said framework is the HAGA, which stands for Hybrid Amber Graph Attention and improves medical image classification by offering a better execution in terms of performance, computation time and, comprehensibility. For these reasons, it can be regarded as a rather prospective tool capable of enhancing the diagnostic outcomes and organizational aspects of MI tasks. As seen in the evaluation of various medical imaging tasks HAGA shows better results but there are few areas of improvement. Adding one or more modality or other clinical data to the existing HAGA and further extending HAGA for seamless integration of multiple imaging could also add more value to diagnosis.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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